

Solving an Inverse Heat Transfer Problem using Physics Informed Neural Networks (PINNs)

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1 Introduction and Goal

Physics Informed Neural Networks (PINNs) are Neural Networks with modified loss function to incorporate physics of problem (by using PDE residuals in loss function). They perform relatively better and requires less data to train. However, added loss terms increases the complexity of loss optimization, making PINNs harder to converge than standard NNs.

One significant advantage of PINNs over standard NNs is that they can be used to solve Inverse problems.

Inverse problems are the problems where we are also required to estimate parameters of a system rather than just surrogate/forward modelling it. These are ill-posed problems as they do not have any single 'correct' solution. In Inverse Heat Transfer Problems, these parameters are Thermal diffusivity, boundary heat flux etc.

In this paper, we recreate M. Billah et. al.[1] paper , in which authors demonstrated solving a specific Inverse Heat Transfer Problem using PINN. Then, We present a better Convergence strategy using Learning Rate and Penalty schedulers. We then compare both methods and conclude our findings.

2 Problem description

The paper describes the 1D Heat Transfer Problem as

Consider a 1D Slab, which receives heat flux $q(t)$ from left side and right side is insulated. Temperature at any position x at time t is $T(x, t)$. Then we can model equations as

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} \quad \text{(Governing Equation) (1)}$$

$$T(x, 0) = 0 \quad \text{(Initial Condition) (2)}$$

$$-k \left. \frac{\partial T}{\partial x} \right|_{x=0} = q(t) \quad \text{(Left B.C.) (3)}$$

$$-k \left. \frac{\partial T}{\partial x} \right|_{x=L} = 0 \quad \text{(Right B.C.) (4)}$$

where L is Length of Slab and α is Thermal Diffusivity.

The Authors considered two cases

- Case 1 : $q(t)$ is constant.
- Case 2 : $q(t)$ is time-variant

There, $q(t)$ further had two subcases : linearly varying and sinusoidally varying. Here for each case/subcase, they also considered a noisy variant of this (with 10% noise added to Dataset).

For Simplicity, in this paper we re-number it to three cases as

- Case 1 : $q(t)$ is constant.
- Case 2 : $q(t)$ is linearly varying in t
- Case 3 : $q(t)$ is sinusoidally varying in t

They then find Analytical solution to above equations for each cases using Green's Function. This Analytical solution, then is used to Create Ground Truth (GT) Dataset. This is discussed in detail in next section.

3 Estimation of Undetermined Parameters

NOTE : Despite authors giving ample of information, Some of critical information required for recreation were missed. These includes

- k - value of thermal conductivity
- Initial guess for parameters q and α

While each of above can be estimated, the exact recreation requires exact initial guess values of q and α . Without this, it can not be clearly said when PINN will eventually converge (or even about the certainty of convergence).

The following is the method for estimation of k and q

Estimation of k

Consider the analytical solution for constant case (detailed in next section and proved in Appendix A)

$$T(x, t) = T_0 + \frac{\alpha q_0 t}{kL} + \frac{2q_0}{kL} \lim_{m \rightarrow \infty} \sum_{n=1}^m \left(\frac{\cos(\gamma_n x) [1 - e^{-\alpha \gamma_n^2 t}]}{\gamma_n^2} \right)$$

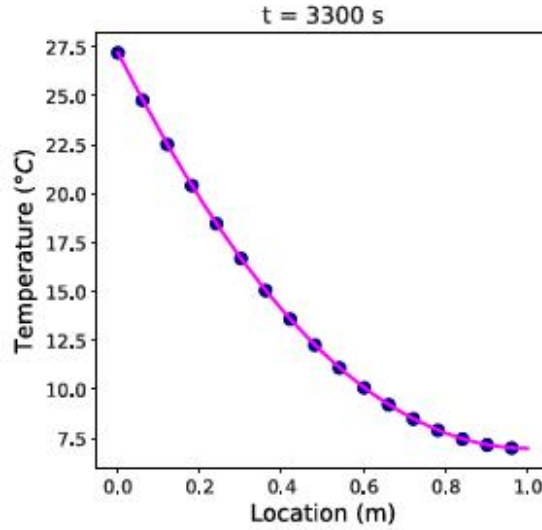


Figure 1: T vs. x Graph at $t = 3300$ s. From the paper

From the T vs. x graph given in paper for constant heat flux case at $t = 3300$ s, we get that $T \approx 27.45^\circ \text{C}$ at $x = 0$

Now, notice that

$$\begin{aligned} e^{-\alpha \gamma_1^2 t} &= e^{-9.868 \times 10^{-5} \times \frac{1\pi^2}{1} \times 3300} = e^{-3.213} = 0.0402 \\ e^{-\alpha \gamma_2^2 t} &= e^{-9.868 \times 10^{-5} \times \frac{4\pi^2}{1} \times 3300} = e^{-12.855} = 2.61 \times 10^{-6} \\ e^{-\alpha \gamma_3^2 t} &= e^{-9.868 \times 10^{-5} \times \frac{9\pi^2}{1} \times 3300} = e^{-28.925} = 2.74 \times 10^{-13} \end{aligned}$$

Hence, from $n = 3$ onwards, we can approximate $1 - e^{-\alpha \gamma_n^2 t} \approx 1$

Now, using above approximation in analytical solution gives

$$T(x, t = 3300) = T_0 + \frac{\alpha q_0 t}{kL} + \frac{2q_0}{kL} \left(\frac{\cos(\gamma_1 x) [1 - e^{-\alpha \gamma_1^2 t}]}{\gamma_1^2} + \frac{\cos(\gamma_2 x) [1 - e^{-\alpha \gamma_2^2 t}]}{\gamma_2^2} + \frac{\cos(\gamma_3 x) [1 - e^{-\alpha \gamma_3^2 t}]}{\gamma_3^2} + \dots \right)$$

$$T(x, t = 3300) = T_0 + \frac{\alpha q_0 t}{kL} + \frac{2q_0}{kL} \left(\frac{\cos(\frac{\pi}{L} x) [1 - 0.0402]}{(\frac{\pi}{L})^2} + \frac{\cos(\frac{2\pi}{L} x) [1 - 2.61 \times 10^{-6}]}{(\frac{2\pi}{L})^2} + \frac{\cos(\frac{3\pi}{L} x)}{(\frac{3\pi}{L})^2} + \dots \right)$$

Putting $x = 0$, and given $L = 1$,

$$T(0, t = 3300) = T_0 + \frac{\alpha q_0 t}{k} + \frac{2q_0}{k} \left(\frac{1}{\pi^2} + \frac{1}{(2\pi)^2} + \frac{1}{(3\pi)^2} + \dots \right) - \frac{2q_0}{k} \left(\frac{0.0402}{\pi^2} + \frac{2.61 \times 10^{-6}}{(2\pi)^2} \right)$$

$$T(0, t = 3300) = T_0 + \frac{\alpha q_0 t}{k} + \frac{2q_0}{k\pi^2} \left(\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right) - \frac{2q_0}{k} \left(\frac{0.0402}{\pi^2} + \frac{2.61 \times 10^{-6}}{(2\pi)^2} \right)$$

Using the fact

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \zeta(2) = \frac{\pi^2}{6}$$

This simplifies to

$$T(0, t = 3300) = T_0 + \frac{\alpha q_0 t}{k} + \frac{2q_0}{k\pi^2} \frac{\pi^2}{6} - \frac{2q_0}{k} \left(\frac{0.0402}{\pi^2} + \frac{2.61 \times 10^{-6}}{(2\pi)^2} \right)$$

$$T(0, t = 3300) = T_0 + \frac{\alpha q_0 t}{k} + \frac{q_0}{3k} - \frac{2q_0}{k} \left(\frac{0.0402}{\pi^2} + \frac{2.61 \times 10^{-6}}{(2\pi)^2} \right)$$

Making k subject

$$k = \frac{\alpha q_0 t + \frac{q_0}{3} - 2q_0 \left(\frac{0.0402}{\pi^2} + \frac{2.61 \times 10^{-6}}{(2\pi)^2} \right)}{T(0, t = 3300) - T_0}$$

Putting

$$T(0, t = 3300) = 27.45$$

$$q_0 = 10,000$$

$$t = 3300 \text{ s}$$

$$\alpha = 9.868 \times 10^{-5}$$

$$T_0 = T(x, 0) = 0$$

This gives

$$k = 237.096$$

Thus, we used $k = 237 \text{ W/mK}$ through out this paper.

Estimation of initial guess for q

We know that

$$q(t) = -k \left. \frac{dT}{dx} \right|_{x=0}$$

Using FDM (specifically Taylor's expansion), we can write

$$\left. \frac{dT}{dx} \right|_{x=0} \approx \left. \frac{dT}{dx} \right|_{x=\Delta x} \approx \frac{T_{x=2\Delta x} - T_{x=0}}{2\Delta x}$$

Though very vague, we just need a initial guess that might help us converge faster.

The values of guesses in different cases are tabulated below

Case	Value(s) in W/mk
Constant (0% Noise)	9461
Constant (10% Noise)	8835
Linear (0% Noise)	[6869 10567 11800 13008 14209 15407 16604 17799 18994]
Linear (10% Noise)	[23499 3956 8680 8713 23774 -22836 32649 10677 45643]
Sinusoidal (0% Noise)	[10321 19442 23572 19919 14467 8929 4903 8599 14070]
Sinusoidal (10% Noise)	[19961 4473 37937 17531 -725 9770 -25054 39899 -8754]

Table 1: Values of initial guesses of q for different cases. For linear and sinusoidal cases, array contains initial guesses for each temporal point, with their decimal part truncated off.

It is clear that for noisy linear and sinusoidal cases, values don't make sense. This is probably due to the fact that only one point at $x=0$ is present for each temporal, unlike constant case where the same was calculated by averaging across 241 different temporal points.

NOTE : These initial guess are only for recreation. For methods in Section 7 and Section 8, initial guess of $q=0$ is taken.

Initial Guess for α

We took initial guess of $\alpha = 0.001$ throughout this experiment (for constant case). Although in general order of thermal diffusivity is about 10^{-4} or 10^{-5} , we took this guess a bit far from true parameter space so as to let it converge.

4 Ground Truth (GT)

Based on the paper,

$$\alpha = 9.868 \times 10^{-5}$$

$$k = 237 \text{ W/mK (estimated)}$$

$$\Delta x = 0.001 \text{ m}$$

$$\Delta t = 1 \text{ s}$$

Case 1 : $q(t)$ is constant

Here, it was given in paper

$$\begin{aligned} q(t) &= q_0 = 10,000 \text{ W/m}^2 \\ t &\in [0, 3600] \text{ seconds} \\ x &\in [0, 1] \text{ m} \end{aligned}$$

It was also stated that for PINN training, 2200 points were randomly chosen out of $48441 = 241 \times 201$ Generated Dataset. This means we first need to generate (241×201) Dataset

Notice that if we generate dataset based on above Δx and Δt , we will have 1001×3600 Data points which is 3.6 Million. Looking at the Numbers from paper, it seems while generation, they used

$$\begin{aligned} \Delta x' &= 5\Delta x = 0.005 \text{ m} \\ \Delta t' &= 15\Delta t = 15 \text{ s} \end{aligned}$$

In this way, we will have $(3600/15 + 1) \times (1/0.005 + 1)$ or 241×201 points

NOTE : This wasn't mentioned in paper, it is purely speculated.

The solution in this case for $q(t) = q_0$ is given as

$$T(x, t) = T_0 + \frac{\alpha q_0 t}{kL} + \frac{2q_0}{kL} \lim_{m \rightarrow \infty} \sum_{n=1}^m \left(\frac{\cos(\gamma_n x) [1 - e^{-\alpha \gamma_n^2 t}]}{\gamma_n^2} \right) \quad (5)$$

where

- $T_0 = T(x, 0) = 0$
- $\gamma_n = \frac{n\pi}{L}$ for $n = 0, 1, 2, 3, \dots$

As we can't take m as ∞ , we take m as 500. To see what minimum value of m should be taken that approximates well, see Appendix B

Case 2 : $q(t)$ is linearly varying in t

Here,

$$\begin{aligned} q(t) &= 10000 + 5t \quad \text{W/m}^2 \\ t &\in [0, 2000] \text{ seconds} \\ x &\in [0, 1] \text{ m} \end{aligned}$$

In paper, it was not given exactly how many datapoints were generated for this case, so we just assume as follows :

As x is same we repeat everything for x i.e. $\Delta x' = 5\Delta x = 0.005$ m, We will have $(1/0.005 + 1) = 201$ spatial points

And $\Delta t' = \Delta t = 1$ s, we will have $(2000/1 + 1) = 2001$ temporal points

That means we first need to generate 2001×201 Dataset

The solution in this case for $q(t) = q_0 + q_1 t$, where $q_0 = 10000, q_1 = 5$, is given as

$$T(x, t) = T_0 + \frac{\alpha q_0 t}{kL} + \frac{\alpha q_1 t^2}{2kL} + \frac{2q_0}{kL} \lim_{m \rightarrow \infty} \sum_{n=1}^m \left(\frac{\cos(\gamma_n x) [1 - e^{-\alpha \gamma_n^2 t}]}{\gamma_n^2} \right) + \frac{2q_1}{\alpha kL} \lim_{m \rightarrow \infty} \sum_{n=1}^m \left(\frac{\cos(\gamma_n x) [\alpha \gamma_n^2 t - 1 + e^{-\alpha \gamma_n^2 t}]}{\gamma_n^4} \right) \quad (6)$$

where

- $\gamma_n = \frac{n\pi}{L}$ for $n = 0, 1, 2, 3, \dots$
- $m = 500$

Case 3 : $q(t)$ is sinusoidally varying in t

Here

$$\begin{aligned} q(t) &= 15000 + 10000 \sin(\omega t) \quad \text{W/m}^2 \\ t &\in [0, 1600] \text{ seconds} \\ x &\in [0, 1] \text{ m} \end{aligned}$$

as given in paper that one complete revolution happens in 1600 second, that means $\omega = \frac{2\pi}{T} = \frac{2\pi}{1600} = 0.003927$

Again, in paper it was not given exactly how many datapoints were generated for this case, so we just took 2001×201 just like in case 2.

The solution in this case for $q(t) = q_0 + q_1 \sin(\omega t)$, where $q_0 = 15000, q_1 = 10000$, is given as

$$T(x, t) = T_0 + \frac{\alpha q_0 t}{kL} + \frac{\alpha q_1}{kL} + \frac{2q_0}{kL} \lim_{m \rightarrow \infty} \sum_{n=1}^m \left(\frac{\cos(\gamma_n x) [1 - e^{-\alpha \gamma_n^2 t}]}{\gamma_n^2} \right) + \frac{2\alpha q_1}{kL} \lim_{m \rightarrow \infty} \sum_{n=1}^m \left(\frac{\cos(\gamma_n x) [\alpha \gamma_n^2 \sin(\omega t) - \omega \cos(\omega t) + \omega e^{-\alpha \gamma_n^2 t}]}{(\alpha \gamma_n^2)^2 + \omega^2} \right) \quad (7)$$

where

- $\gamma_n = \frac{n\pi}{L}$ for $n = 0, 1, 2, 3, \dots$
- $m = 500$

The Proofs of these equations are given in Appendix A

5 Data Extraction for PINN

5.1 Constant Case

For constant case, 2200 points were randomly chosen from 241×201 points.

i.e. PINN Input data for this case was a 2200 length array of tuples (x_i, t_j, T_{ij})

5.2 Linear Case

For linear case, 201 points were chosen from 2001×201 points for each temporal point $[1, 250, 500, 750, 1000, 1250, 1500, 1750, 2000]$ seconds

i.e. an array of tuples (x_i, t_j, T_{ij}) of length $201 \times 9 = 1809$

The reason for choosing $t = 1$ s instead of taking $t = 0$ s will be explained later.

5.3 Sinusoidal Case

For Sinusoidal, 201 points were chosen from 1601×201 points for each temporal point $[1, 160, 400, 640, 800, 960, 1200, 1440, 1600]$ seconds

i.e. an array of tuples (x_i, t_j, T_{ij}) of length $201 \times 9 = 1809$

6 Method Overview for Recreation

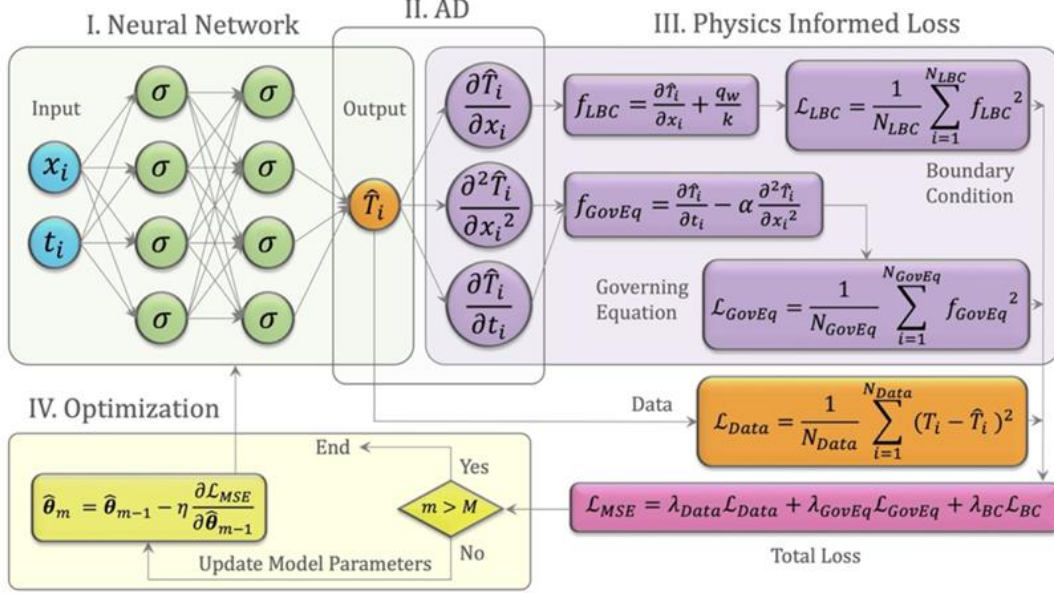


Figure 2: Image showing workflow for their method. Taken from the paper

Here, parameters $\hat{\theta}_m = [\theta_1 \theta_2 \dots \theta_n q \alpha]_m$, where

- m denotes the step
- $\theta_1 \theta_2 \dots \theta_n$ are neural network parameters
- q and α are thermophysical parameters (which are to be estimated)

Let $\hat{\theta}_{NN} = [\theta_1 \theta_2 \dots \theta_n]$, then

$$\begin{aligned}
 (\hat{\theta}_{NN})_m &= (\hat{\theta}_{NN})_{m-1} - \eta_{NN} \frac{\partial \mathcal{L}_{MSE}}{\partial (\hat{\theta}_{NN})_{m-1}} \\
 q_m &= q_{m-1} - \eta_q \frac{\partial \mathcal{L}_{MSE}}{\partial q_{m-1}} \\
 \alpha_m &= \alpha_{m-1} - \eta_\alpha \frac{\partial \mathcal{L}_{MSE}}{\partial \alpha_{m-1}}
 \end{aligned} \tag{8}$$

where

$$\mathcal{L}_{MSE} = \lambda_{data} \mathcal{L}_{data} + \lambda_{gov} \mathcal{L}_{gov} + \lambda_{lbc} \mathcal{L}_{lbc} \tag{9}$$

Here, authors considered $\eta_{NN} = \eta_\alpha = \eta_q = 0.001$

and penalty term weights $(\lambda_{data}, \lambda_{lbc}, \lambda_{gov})$ are considered as follows

Case	λ_{data}	λ_{gov}	λ_{lbc}
Constant	10	10	1
Linear	1000	—	1
Sinusoidal	1000	—	1

Table 2: Penalty term weights taken for each case. For linear and sinusoidal cases, governing equation term was discarded

Model architecture parameters are tabulated below

Case	Number of layers	Number of Neurons per layer	Activation function
Constant	8	20	tanh
Linear	4	20	tanh
Sinusoidal	4	20	tanh

Table 3: Model Architecture for each case as per this method

Data Scaling

For linear and sinusoidal cases, the values of temperature and time were scaled using min-max scaling as

$$T' = \frac{T - T_{min}}{T_{max} - T_{min}} \quad \& \quad t' = \frac{t - t_{min}}{t_{max} - t_{min}} \quad (10)$$

Here, $t_{min} = 0$ and $T_{min} = 0$

It can be shown from differential equations ((1) and (3)) that this follows :

$$q' = \frac{q}{T_{max}} \quad \& \quad \alpha' = \alpha \times t_{max}$$

The value of positions were not scaled as it is already between 0 and 1.

For constant case, only time was scaled and not Temperature.

Results of recreation are given on next page :

6.1 Results of Recreation

6.1.1 Constant Heat Flux

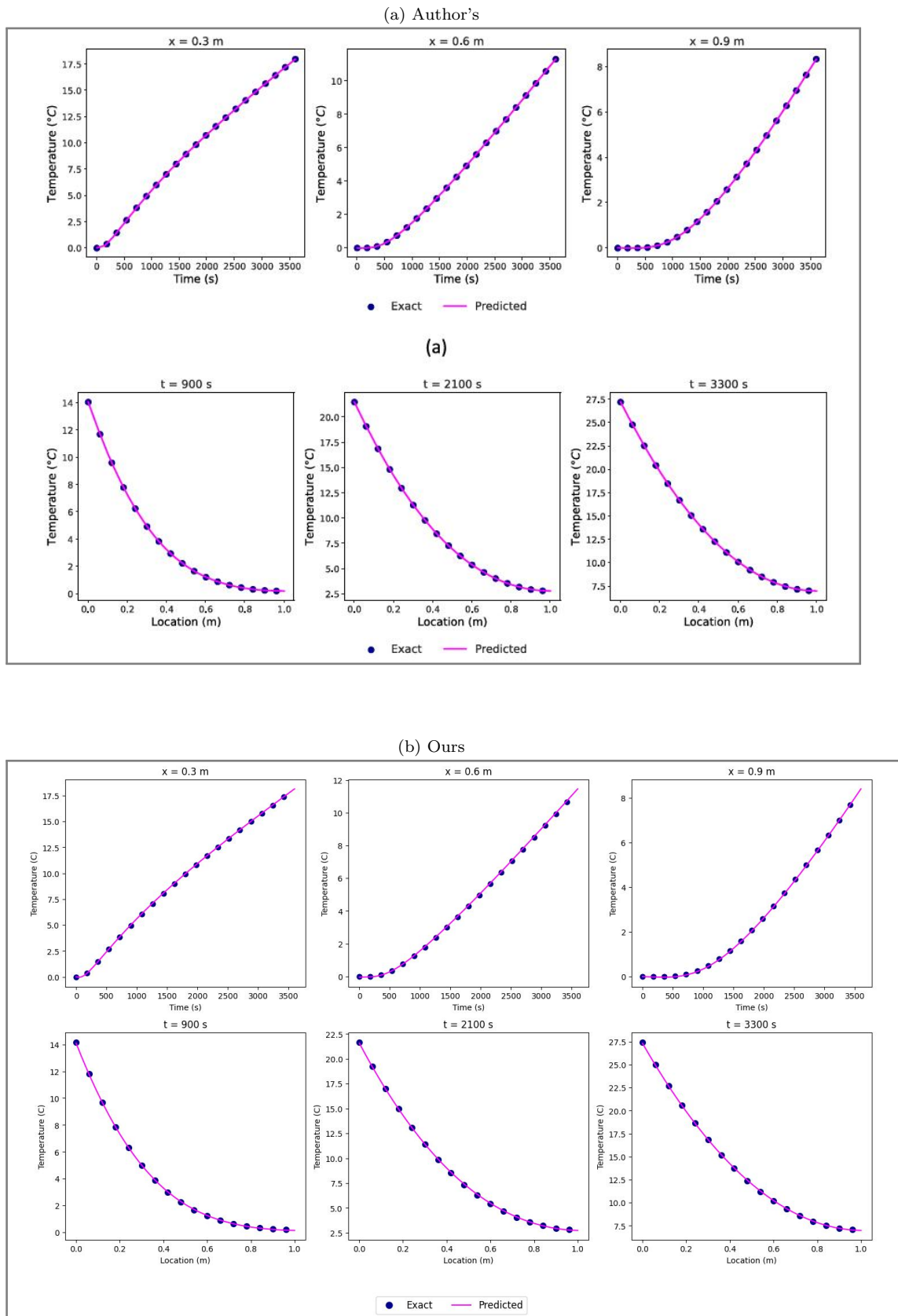


Figure 3: Temperature profile for Constant Heat Flux case (without Noise)

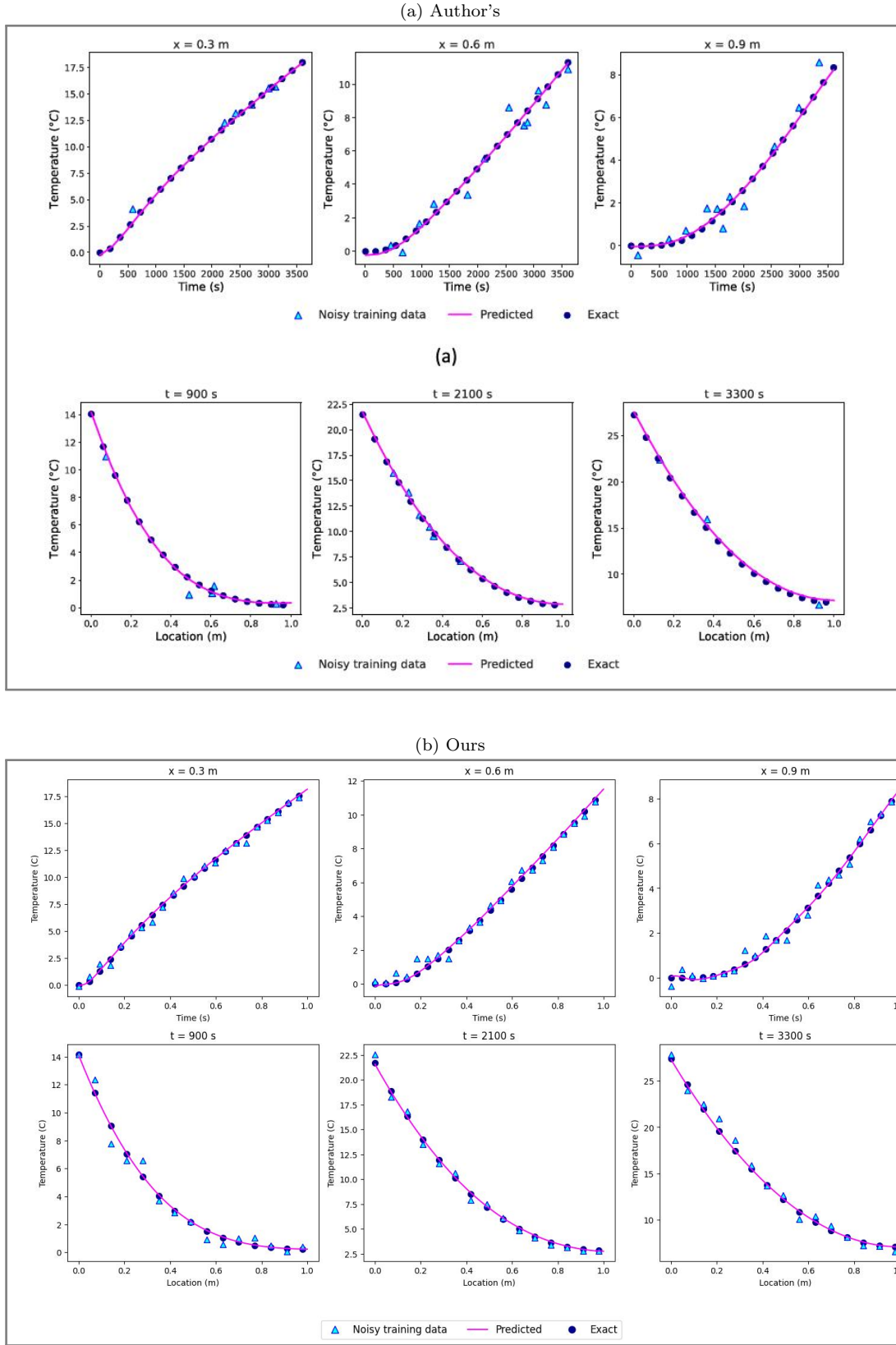


Figure 4: Temperature profile for Constant Heat Flux case (10% Noise)

	α (m^2/s) (author's)	q (W/m^2) (author's)	α (m^2/s) (ours)	q (W/m^2) (ours)
True Value	9.868e-5	10000	9.868e-5	10000
Predicted Value (0% Noise)	9.895e-5	9989	10.170e-5	9798
Predicted Value (10% Noise)	9.854e-5	9968	10.237e-5	9774

Table 4: Comparison of thermophysical parameters predicted by model recreating their experiment

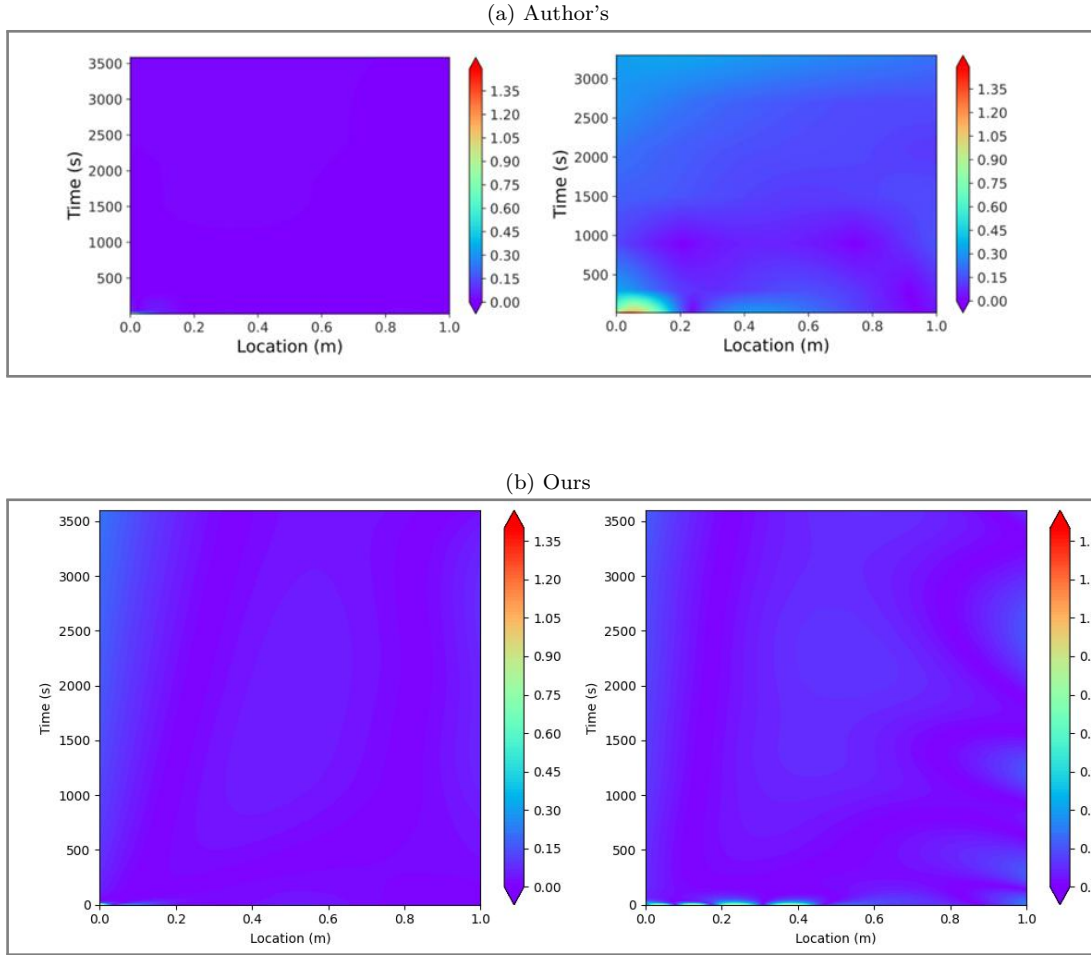


Figure 5: Absolute Error ($^{\circ}\text{C}$) Map. Figure on Left is for 0% Noise Constant Heat Flux case and on right is for 10% Noise Constant Heat Flux Case

Time taken	Author's (hours)	Ours (hours)
Constant case (0% Noise)	11.2*	12
Constant case (10% Noise)	11.2*	12

* It wasn't exactly stated how much time they took for this case, based on 2000 point training time given in table in their paper, this is a estimate

Table 5: Comparison of time taken in Constant Heat Flux Case

6.1.2 Linear Heat Flux

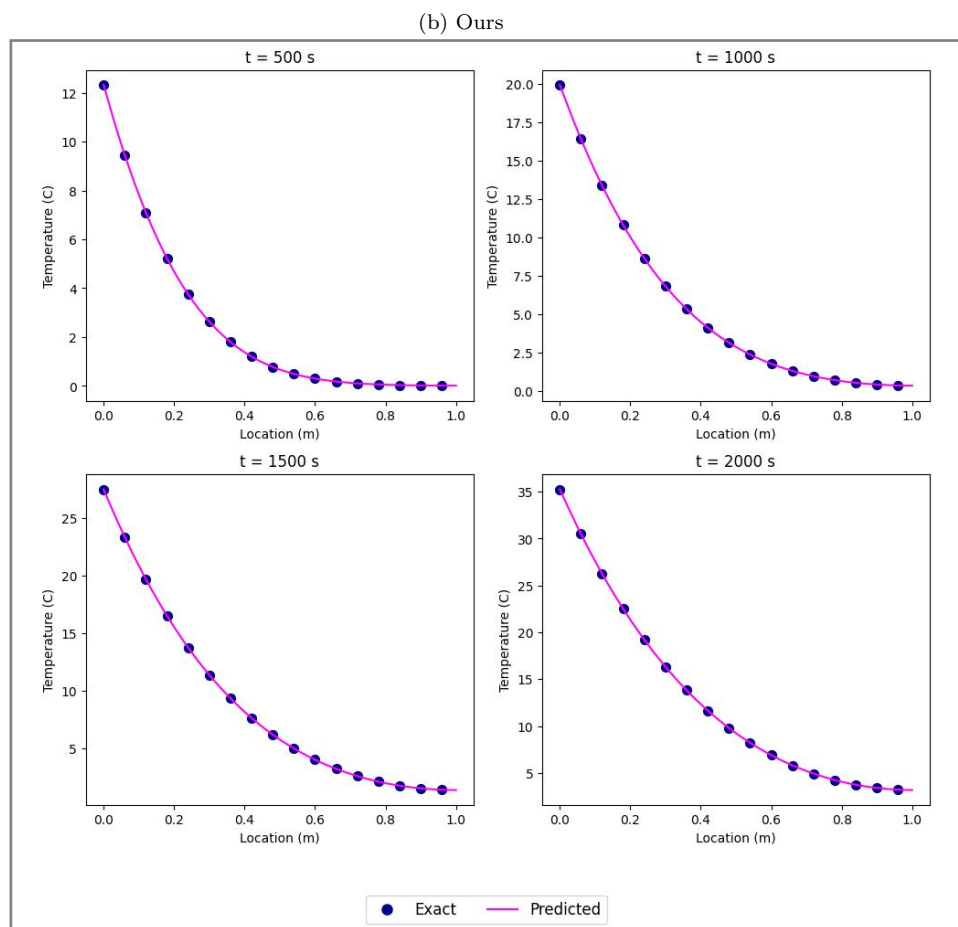
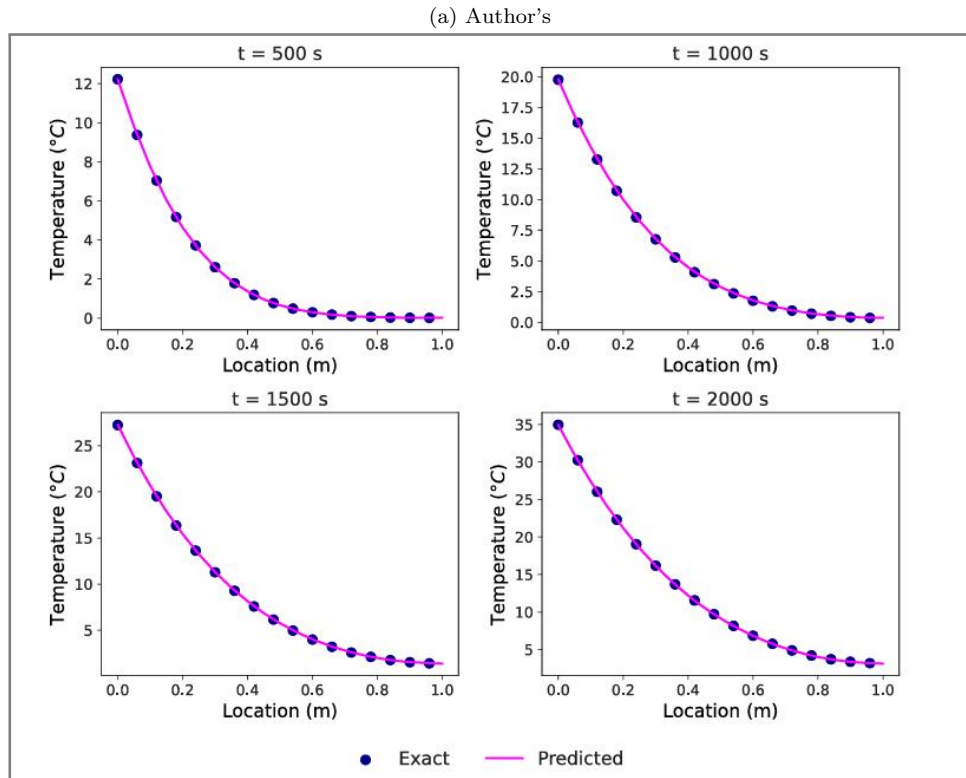
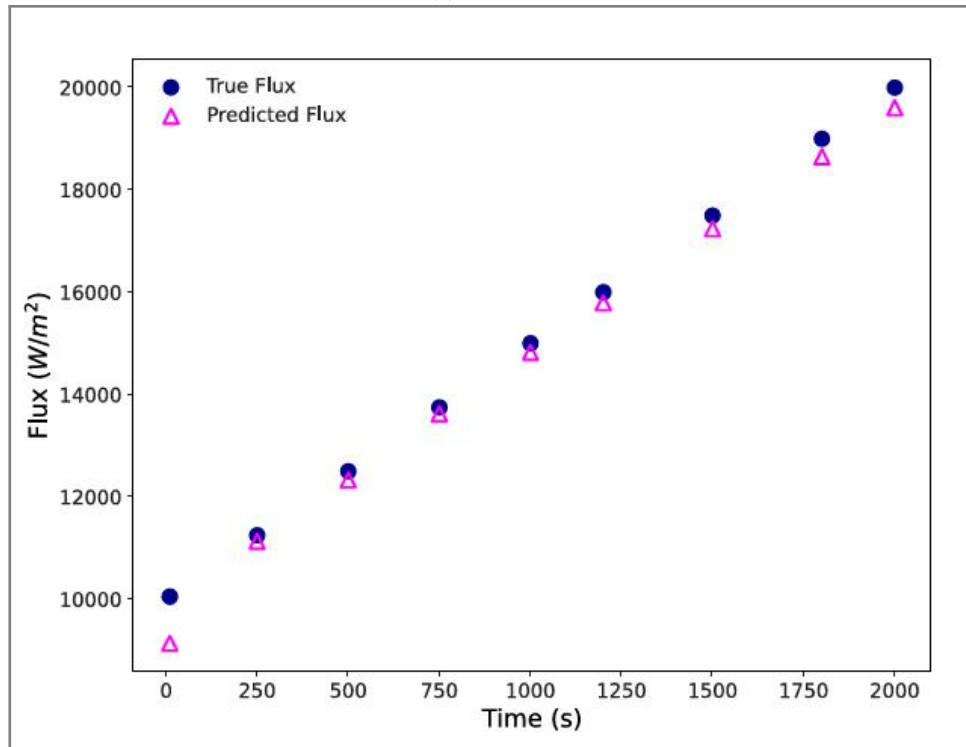


Figure 6: Temperature profile for Linear Heat Flux case (0% Noise)

(a) Author's



(b) Ours

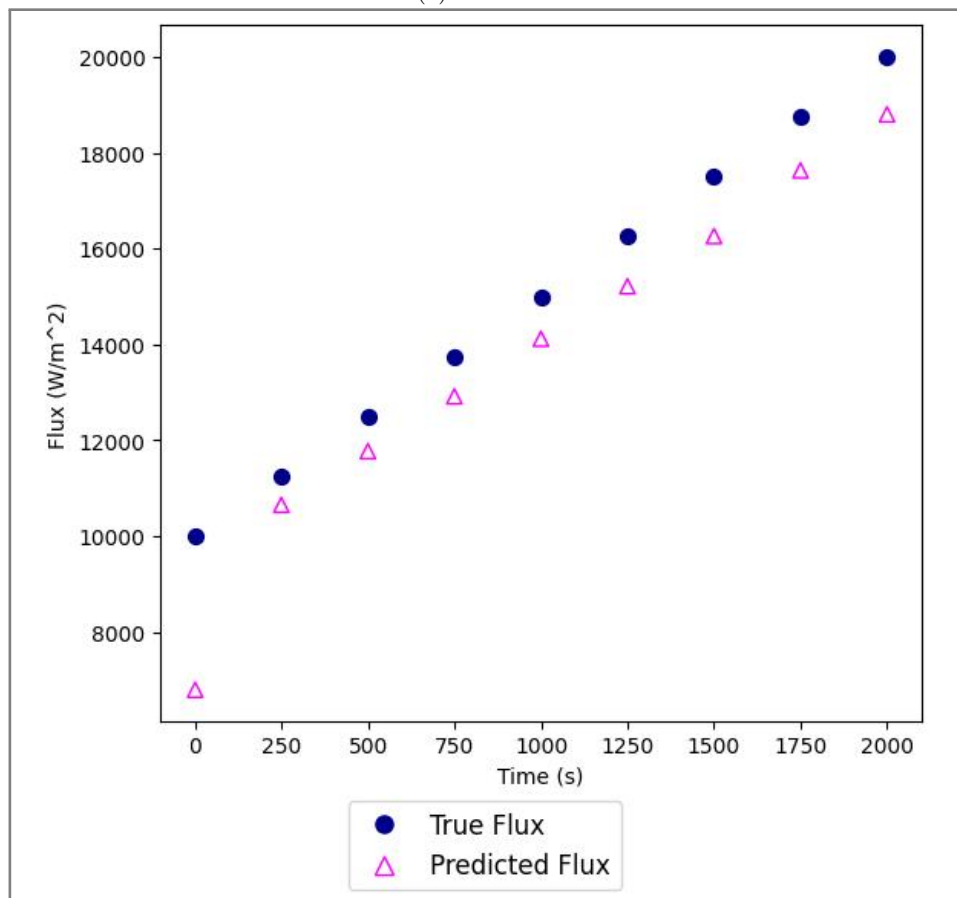


Figure 7: Predicted Heat Flux for Linear Heat Flux case (0% Noise)

More in progress ...

6.1.3 Sinusoidal Heat Flux

In progress ...

7 Improved Training for Faster Convergence

We made the following changes

- For constant case, in equations (8), we chose learning rates $\eta_q = 5$, $\eta_{NN} = \eta_\alpha = 0.001$ instead of choosing all as 0.001. Penalty term weights are $\lambda_{data} = 20$, $\lambda_{gov} = 10$ $\lambda_{lbc} = 1$.
- For linear and sinusoidal case, in equations (8), we scheduled learning rates as

$$\eta_{NN} = \begin{cases} 0.001 & \text{if } \text{epochs} \leq 20,000 \\ 0 & \text{if } 20,000 < \text{epochs} \leq 30,000 \\ 0.001 \times 0.05 = 0.00005 & \text{if } 30,000 < \text{epochs} \leq 70,000 \end{cases}$$

$$\eta_q = \begin{cases} 6.5 & \text{if } \text{epochs} \leq 40,000 \\ 6.5 \times 0.8 = 5.2 & \text{if } 40,000 < \text{epochs} \leq 70,000 \end{cases}$$

and penalty term weight schedulers as

$$\lambda_{lbc} = \begin{cases} 0, & \text{if } \text{epochs} \leq 20,000 \\ 1, & \text{if } 20,000 < \text{epochs} \leq 70,000 \end{cases}$$

where $\lambda_{data} = 1000$. Governing equation term was discarded for these cases.

- Initial guess of $q = 0$ for every case
- The following architecture was used

Case	Number of layers	Number of Neurons per layer	Activation function
Constant	8	20	tanh
Linear	3	30	SiLU
Sinusoidal	3	30	SiLU

Table 6: Model Architecture for each case as per our method

Data Scaling was same as in Section 6

Results are shown on next page :

7.1 Results on Improved Training

7.1.1 Constant Heat Flux

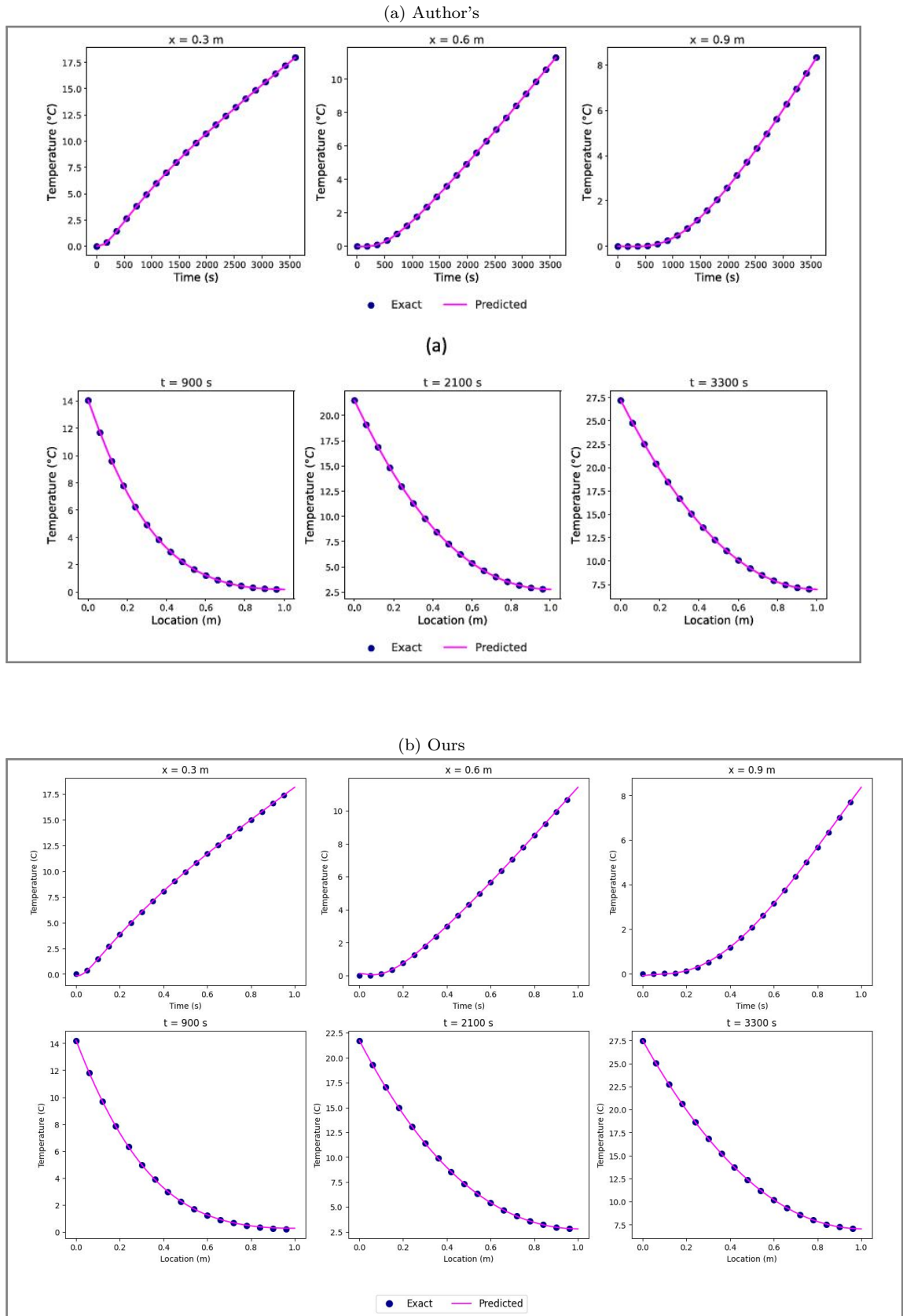


Figure 8: Temperature profile for Constant Heat Flux case (0% Noise)

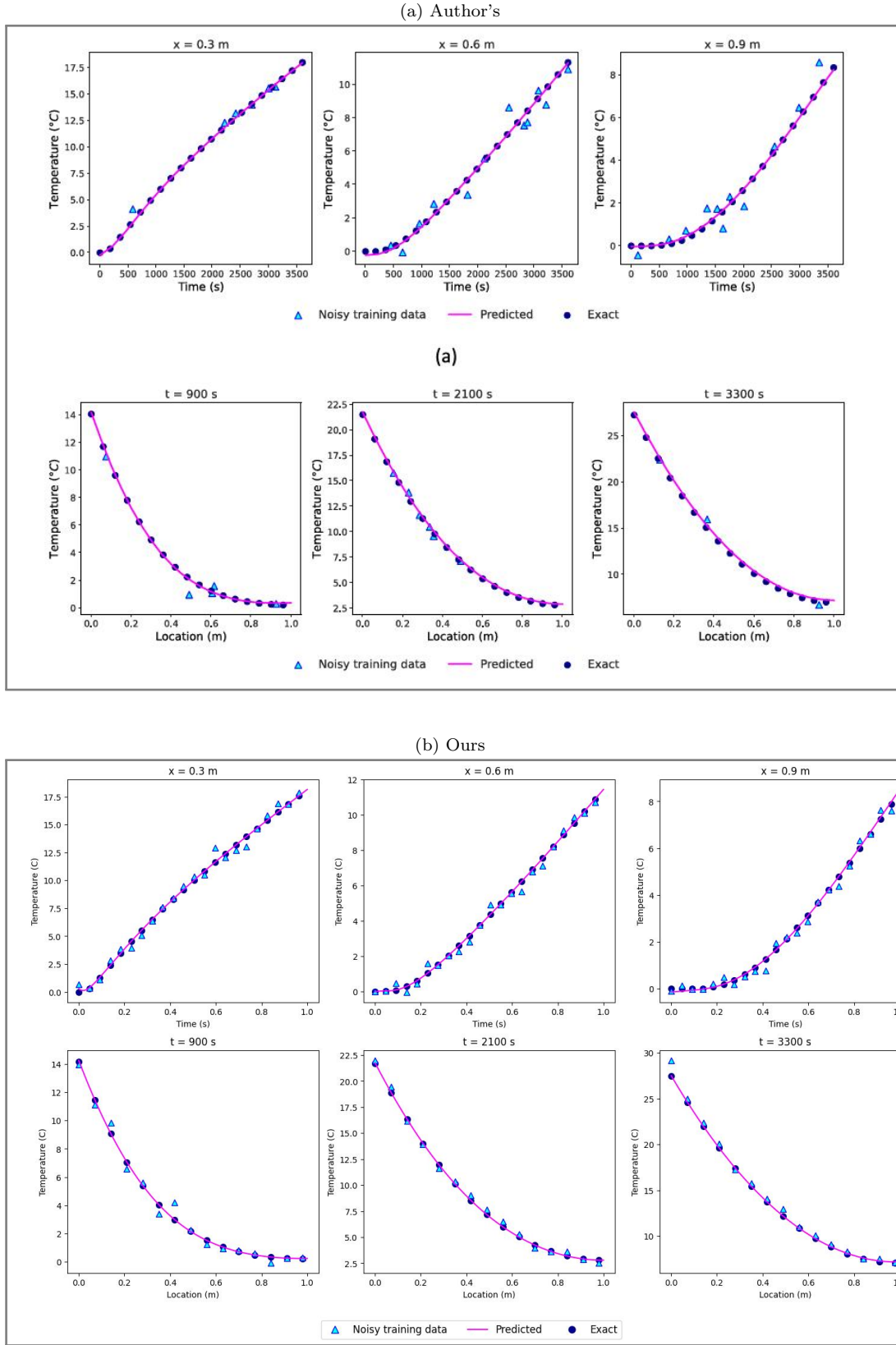


Figure 9: Temperature profile for Constant Heat Flux case (10% Noise)

	α (m^2/s) (author's)	q (W/m^2) (author's)	α (m^2/s) (ours)	q (W/m^2) (ours)
True Value	$9.868e-5$	10000	$9.868e-5$	10000
Predicted Value (0% Noise)	$9.895e-5$	9989	$9.862e-5$	9999.7
Predicted Value (10% Noise)	$9.854e-5$	9968	$9.790e-5$	10042

Table 7: Comparison of thermophysical parameters predicted by model trained using improved method

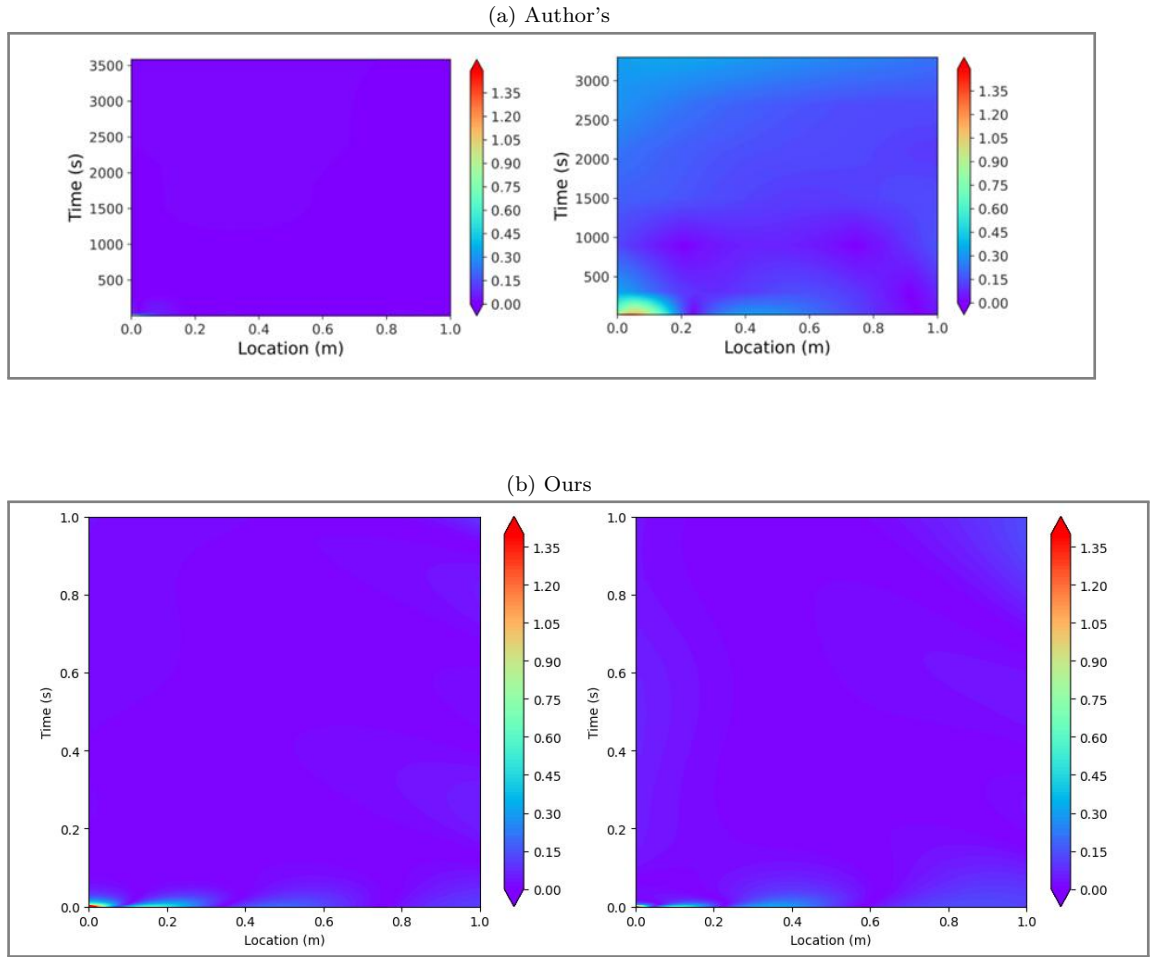


Figure 10: Absolute Error ($^{\circ}C$) Map. Figure on Left is for 0% Noise Constant Heat Flux case and on right is for 10% Noise Constant Heat Flux Case

Time taken	Author's (hours)	Ours (hours)
Constant case (0% Noise)	11.2*	0.53
Constant case (10% Noise)	11.2*	0.48

* It wasn't exactly stated how much time they took for this case, based on 2000 point training time given in table in their paper, this is a estimate

Table 8: Comparision of time taken for constant Heat Flux case, trained with improved method.

7.1.2 Linear Heat Flux

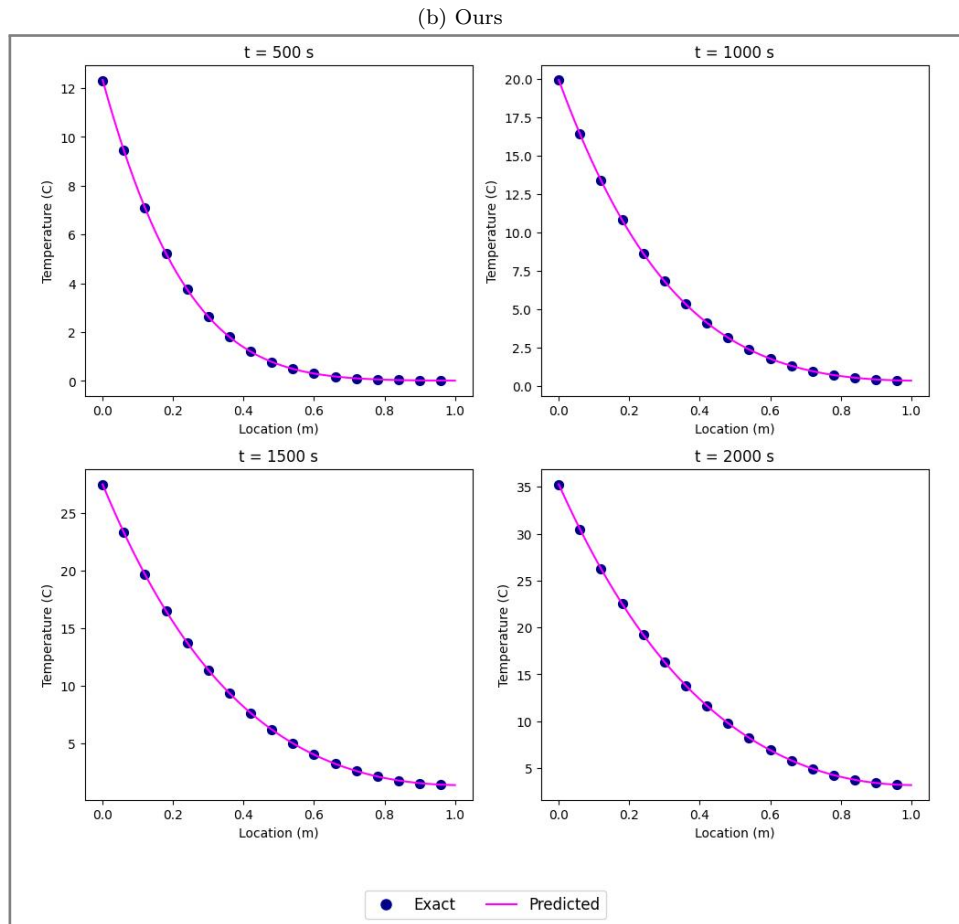
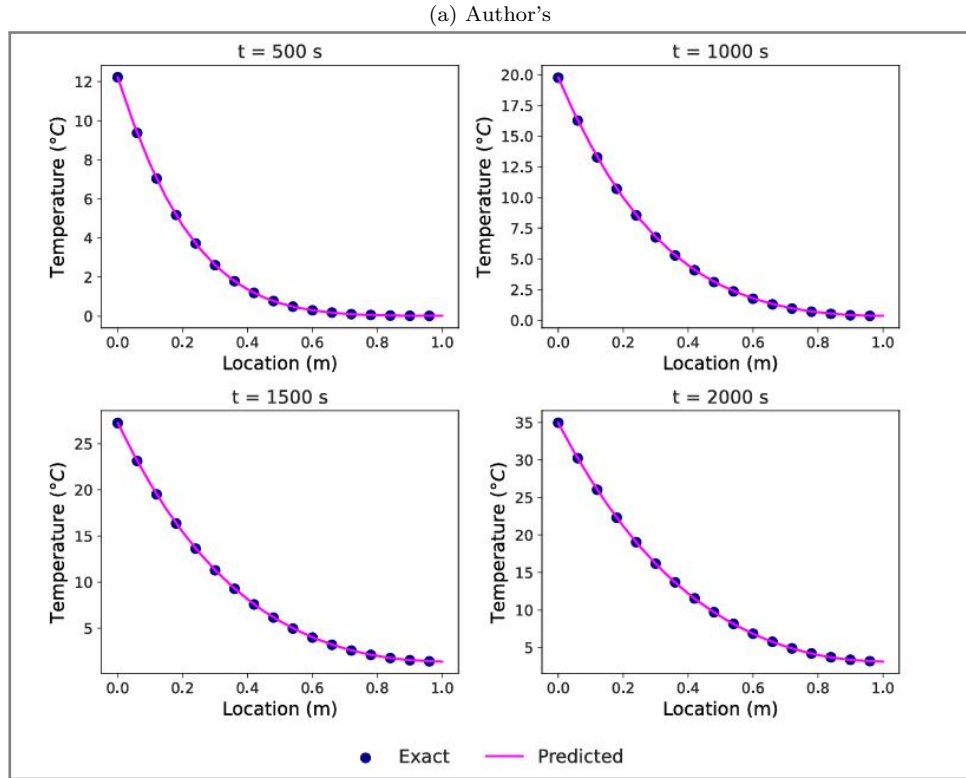
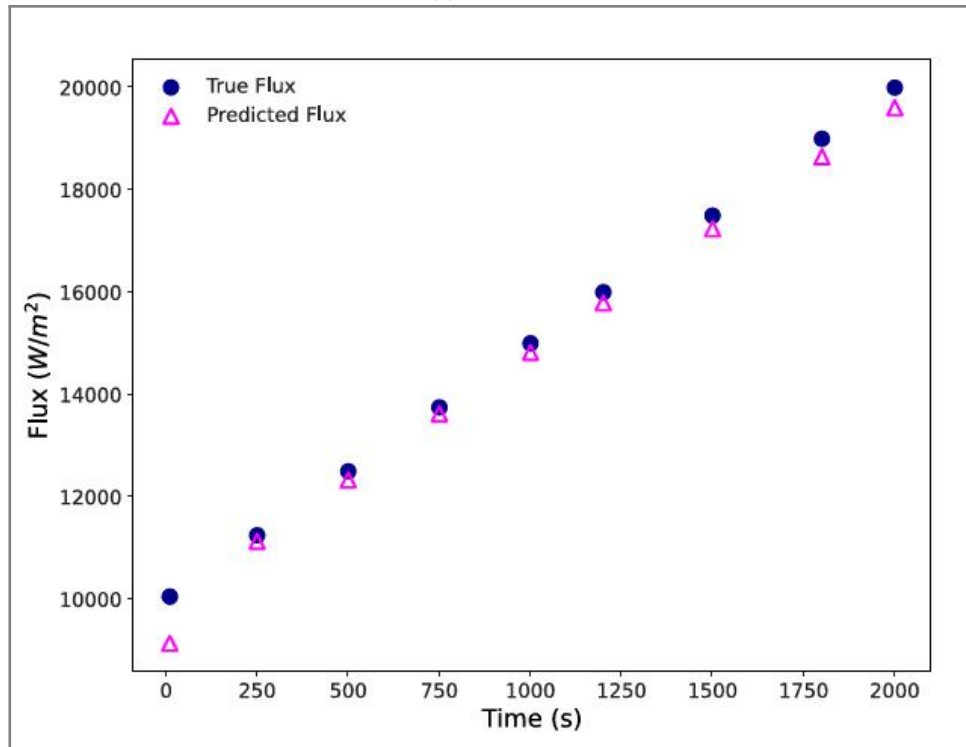


Figure 11: Temperature profile for Linear Heat Flux case (0% Noise)

(a) Author's



(b) Ours

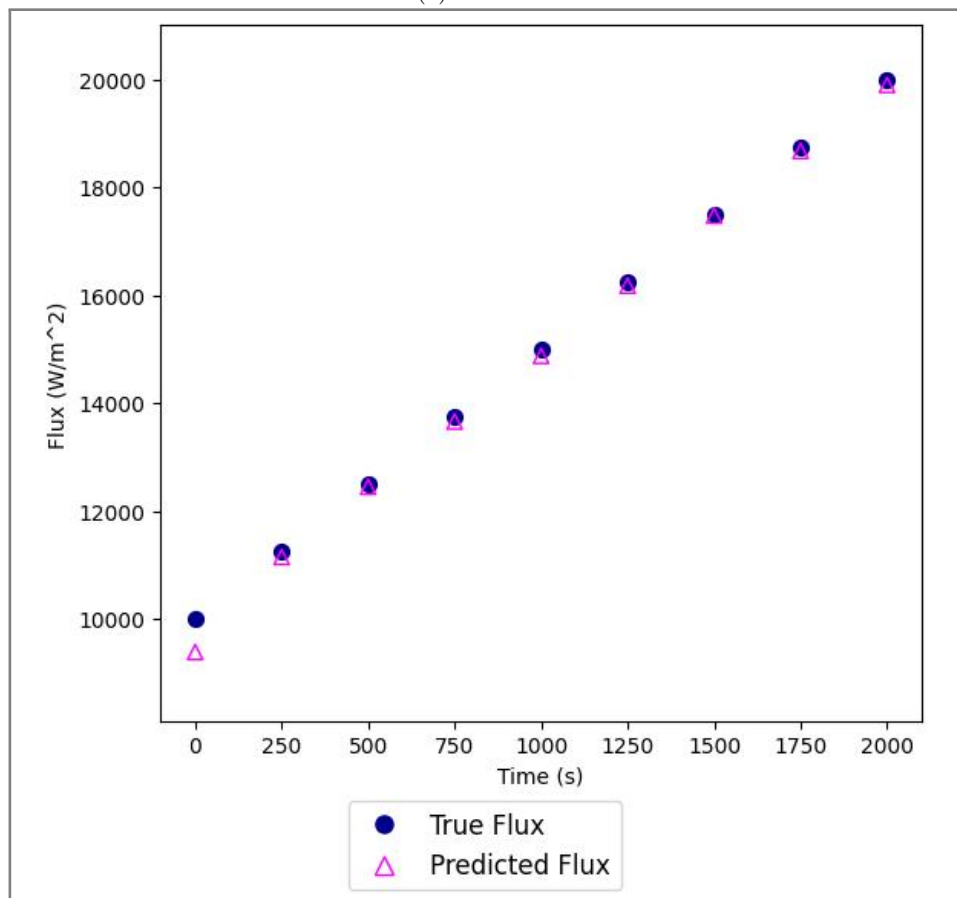
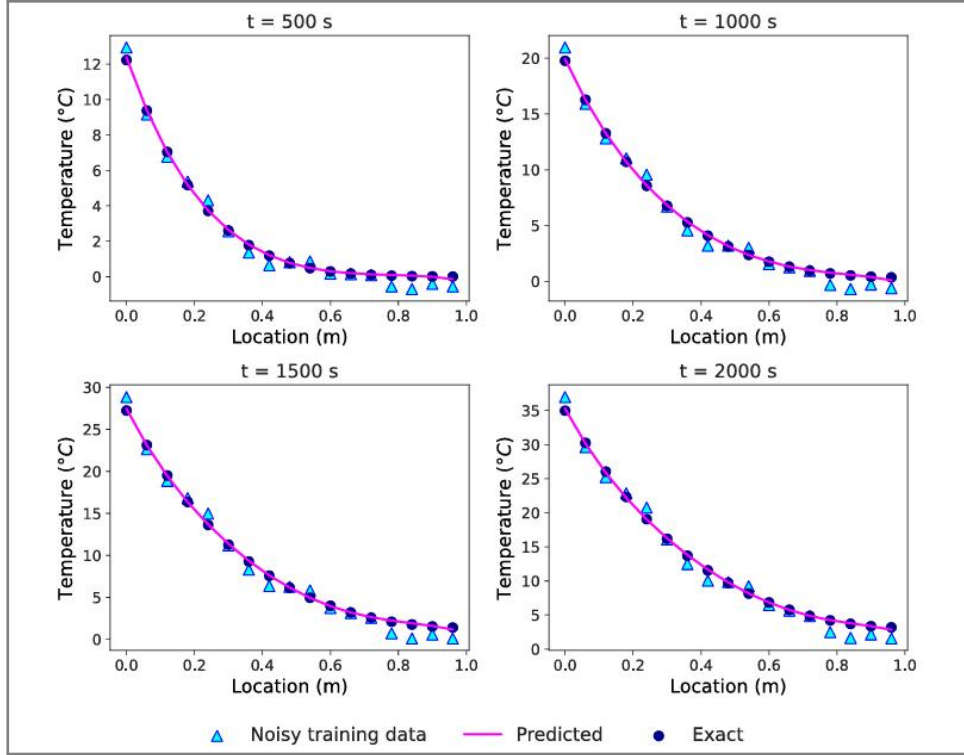


Figure 12: Predicted Heat Flux for Linear Heat Flux case (0% Noise)

(a) Author's



(b) Ours

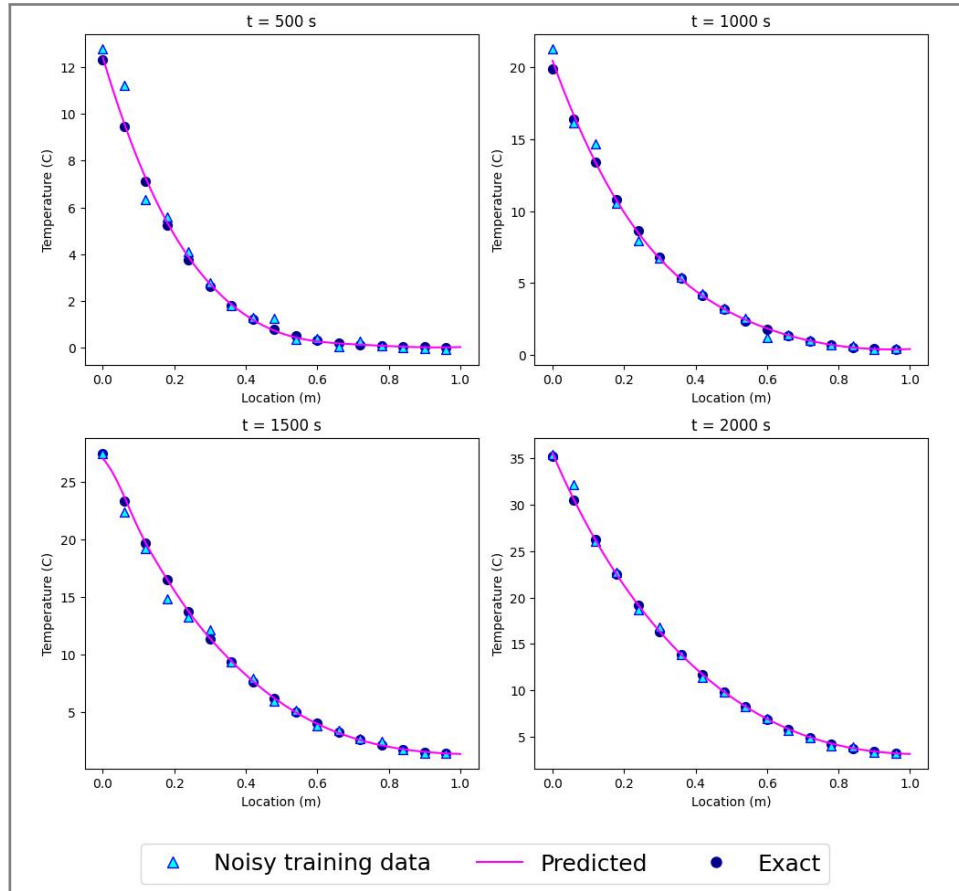


Figure 13: Temperature profile for Linear Heat Flux case (10% Noise)

Time taken	Author's (hours)	Ours (hours)
Linear case (0% Noise)	N/A ^a	1.5
Linear case (10% Noise)	N/A ^a	1.5

^a It wasn't stated how much time they took for this case.

Table 9: Comparison of time taken for Linear Heat Flux case, trained with improved method.

7.1.3 Sinusoidal Heat Flux

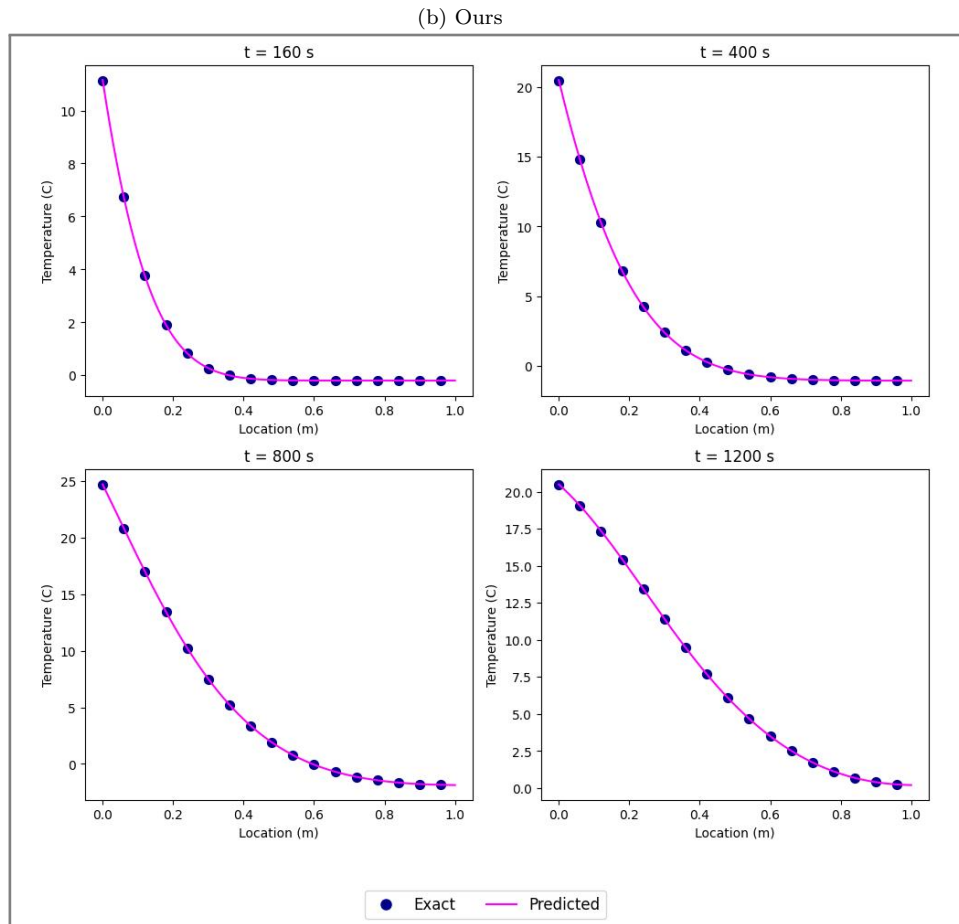
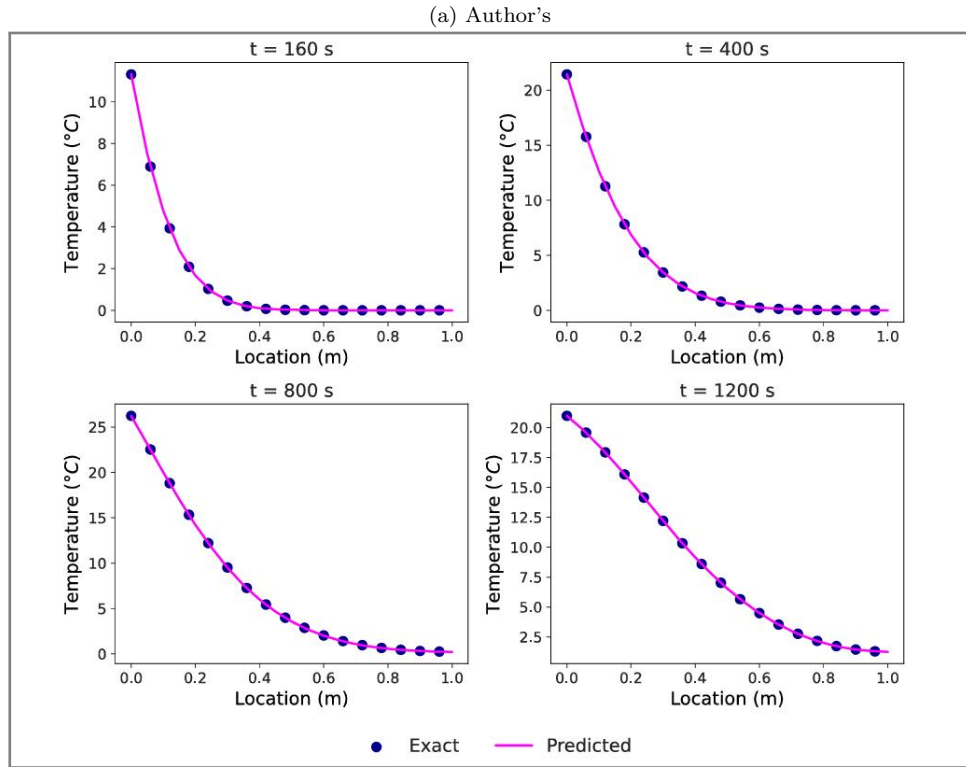
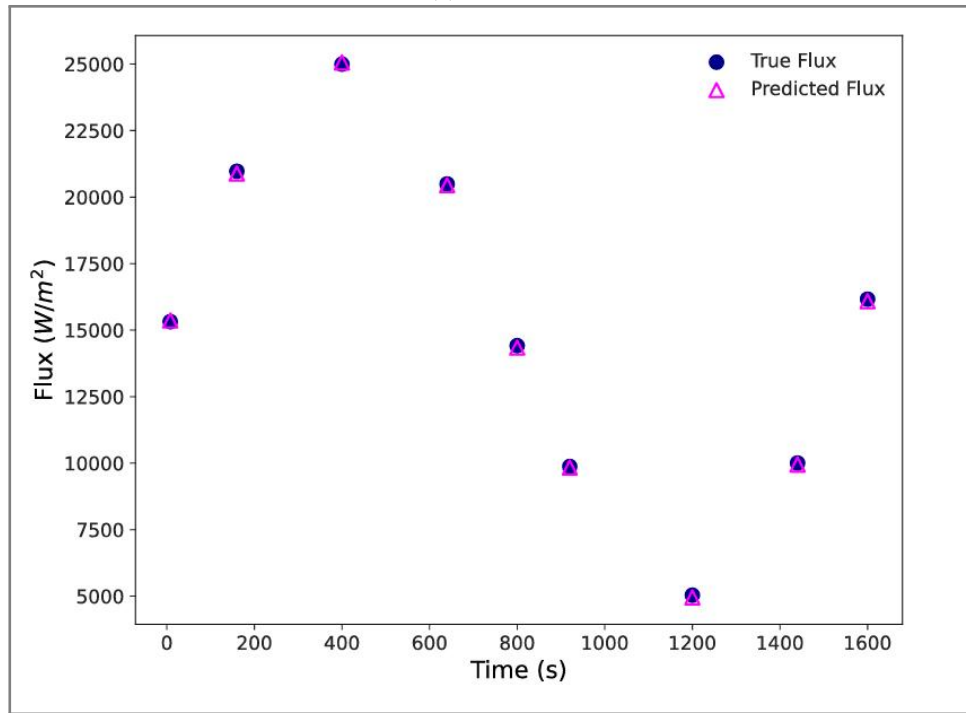


Figure 14: Temperature profile for Sinusoidal Heat Flux case (0% Noise)

(a) Author's



(b) Ours

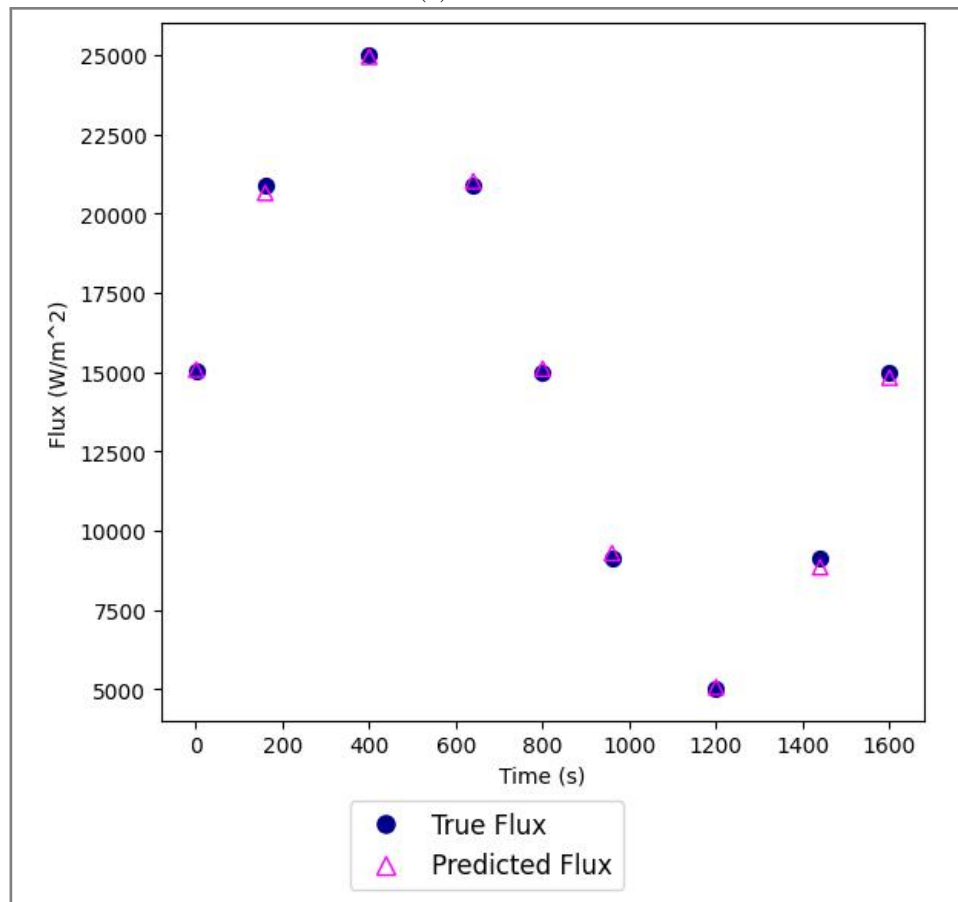
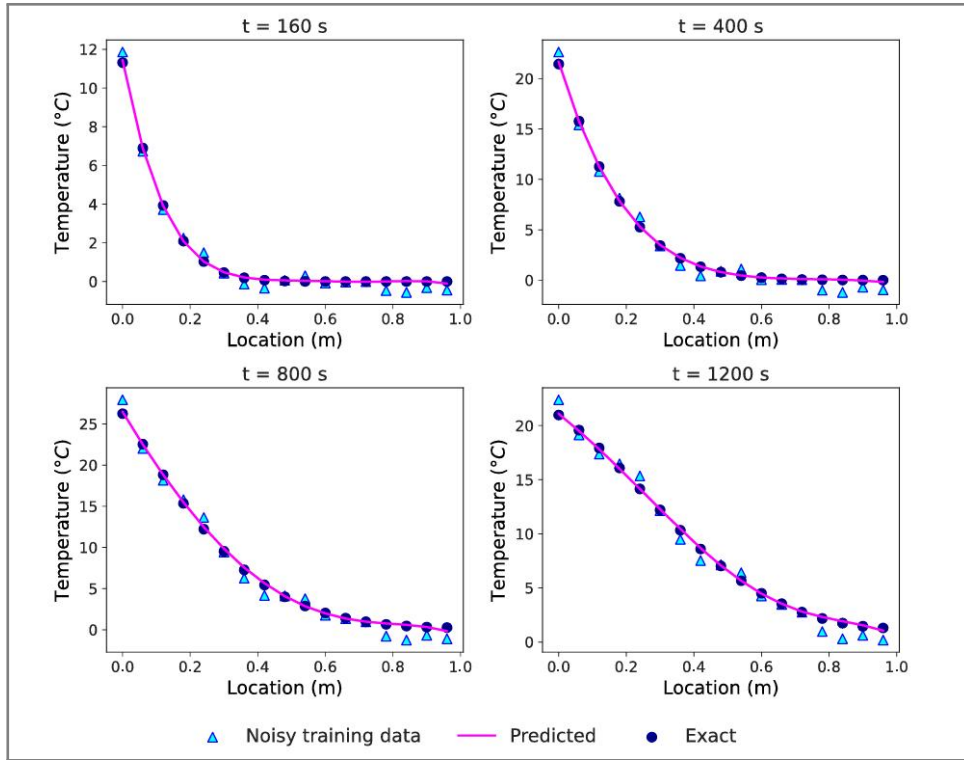


Figure 15: Predicted Heat Flux for Sinusoidal Heat Flux case (0% Noise)

(a) Author's



(b) Ours

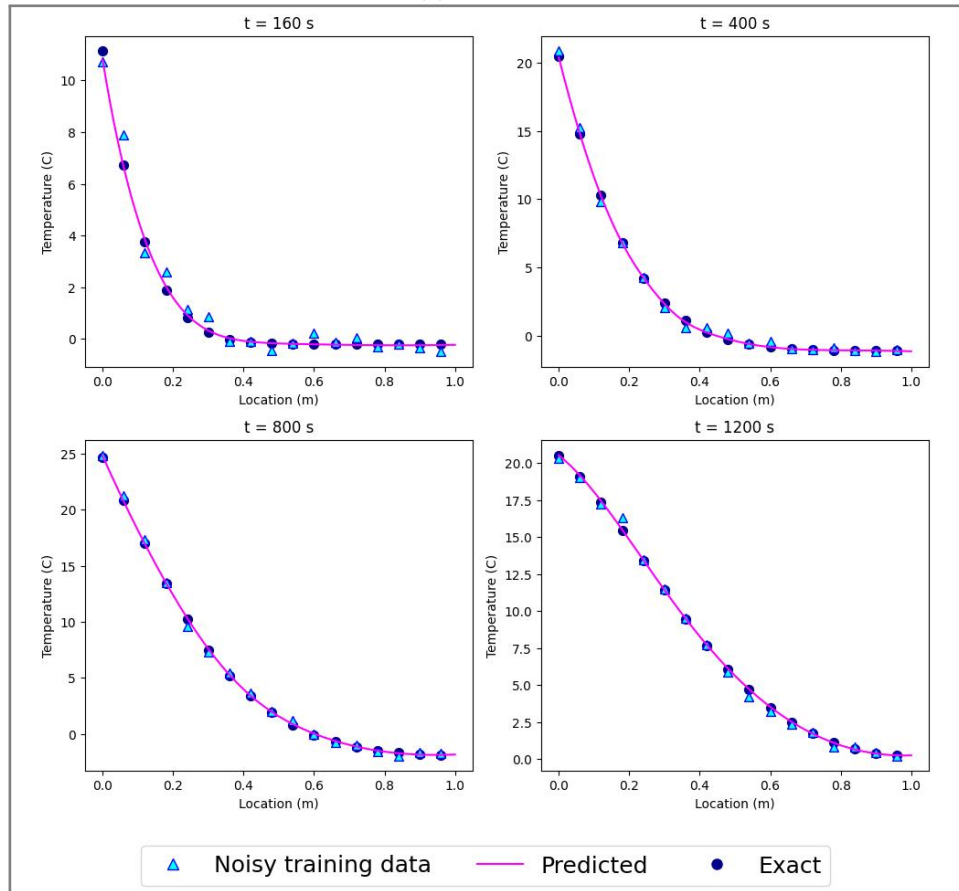


Figure 16: Temperature profile for Constant Heat Flux case (10% Noise)

Time taken	Author's (hours)	Ours (hours)
Sinusoidal case (0% Noise)	N/A ^a	1.5
Sinusoidal case (10% Noise)	N/A ^a	1.5

^a It wasn't stated how much time they took for this case.

Table 10: Comparison of time taken for Sinusoidal Heat Flux case, trained with improved method.

References

- [1] Md Muhtasim Billah, Aminul Islam Khan, Jin Liu, and Prashanta Dutta. Physics-informed deep neural network for inverse heat transfer problems in materials. *Materials Today Communications*, 35:106336, 2023.